29/9/25

LOTUS

$$E[g(X)] = \sum_x g(x) p_X(x) \quad \text{(or)} \quad \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

$Y = g(X)$

$$p_Y(y) = \sum_{x: g(x)=y} p_X(x)$$

$$E[Y] = \sum_y y\, p_Y(y) = \sum_y y \sum_{x: g(x)=y} p_X(x) = \sum_x g(x) p_X(x)$$

$X_1, X_2, \ldots, X_n \sim$ joint distribution

$$E[g(X_1, \ldots, X_n)] = \sum_{x^n} g(x^n) p_{X^n}(x^n)$$

$g: \mathbb{R}^n \to \mathbb{R}$

$g(x_1, \ldots, x_n)\, p_{X_1 \cdots X_n}(x_1, \ldots, x_n)$

(or)

$$\int \cdots \int_{\mathbb{R}^n} g(x^n) f_{X^n}(x^n)\, dx^n$$

$g(x_1, \ldots, x_n)\, f_{X_1 \cdots X_n}(x_1, \ldots, x_n)\, dx_1 \cdots dx_n$

Linearity Property:

① $E[aX + b] = a E[X] + b$ $\quad E[\text{constant}] = \text{same constant}$

②

$$E[a X_1 + a_2 X_2 + \cdots + a_n X_n + b] = a_1 E[X_1] + a_2 E[X_2] + \cdots + a_n E[X_n] + b$$

any joint distribution

Ex: ① Throw a die twice independently X_1, X_2: two values

$E[X_1] = 3.5 = E[X_2]$ $\quad E[X_1 + X_2] = 7 = \sum_{i,j} (i+j) \frac{1}{36}$

$$E[X_1 X_2] = \sum_i \sum_j ij\, p_{X_1 X_2}(i,j) = \sum_i \sum_j ij\, p_{X_1}(i) p_{X_2}(j) = \left(\sum_i i\, p_{X_1}(i)\right)\left(\sum_j j\, p_{X_2}(j)\right)$$

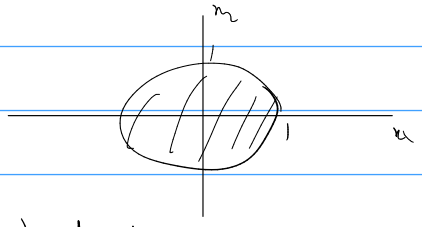
X_1, X_2 indep $\Rightarrow E[g_1(X_1) g_2(X_2)] = E[g_1(X_1)] E[g_2(X_2)]$ indep.

$$E\left[X_1^2 + 3X_1X_2 + X_2^2\right] = 2 \cdot \frac{7\times 13}{6} + 3 \times \frac{7}{2} \times \frac{7}{2}$$

(2) $X_1, X_2 \sim$ iid Uniform $[0,1]$

repeat the above

(3) $X_1, X_2 \sim$ Uniform on unit circle



$$2E[X_1^2] = E\left[X_1^2 + X_2^2\right] = \int_{x_1=-1}^{1} \int_{x_2=-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} (x_1^2 + x_2^2) \cdot \frac{1}{\pi} dx_1 dx_2 \overset{?}{=} 1/2$$

(4) $X_1, X_2 \sim$ iid

$$E\left[\frac{X_1}{X_1+X_2}\right] = E\left[\frac{X_2}{X_1+X_2}\right] = 1/2$$

(5) $X_1, X_2, \ldots, X_n \sim$ iid Bernoulli(p)

$$Y = X_1 + X_2 + \cdots + X_n \sim \text{Binomial}(n,p)$$

$$E[Y] = np \qquad Y = \sum_{i=1}^{n} X_i$$

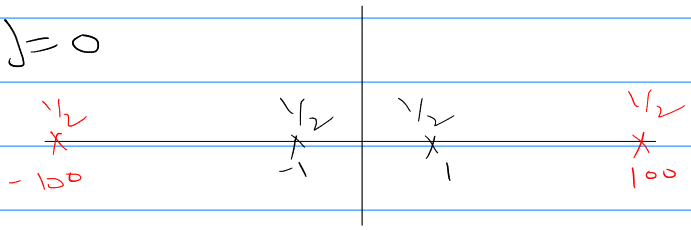
$$E[Y^2] = n \cdot p + n(n-1)p^2 \qquad Y^2 = \sum_i X_i \sum_j X_j$$

$$= n^2p^2 + np(1-p) \qquad = \sum_i \sum_j X_i X_j$$

$$E[X_iX_j] = \begin{cases} p, & i = j \\ p^2, & i \neq j \end{cases}$$

Variance

Say $E[X] = 0$

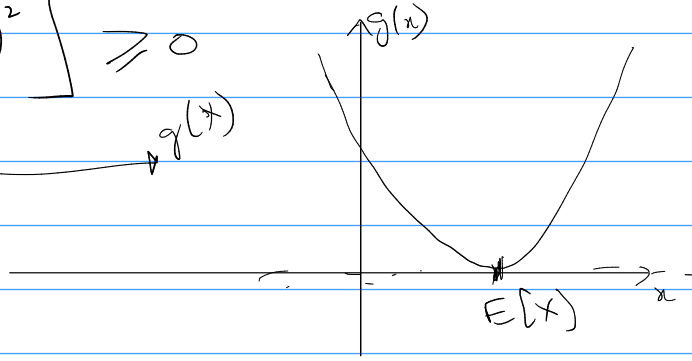


→ Need some measure of spread

Defn: Variance of a RV X is

$$\mathrm{Var}(X) = E\left[(X - E[X])^2\right] \geq 0$$

($E[X]$: const; $g(x)$)



Standard Deviation

$$\sigma(X) = \sqrt{\mathrm{Var}(X)}$$

$$\mathrm{Var}(X) = E\left[X^2 - 2E[X]X + E[X]^2\right]$$

$$= E[X^2] - E[X]^2 \geq 0$$

$$\boxed{E[X^2] \geq E[X]^2}$$

Equality if X = constant w.p. 1

Ex:

(1) $X \sim \mathrm{Bernoulli}(p)$

$E[X] = p \qquad \mathrm{Var}(X) = p - p^2$

(2) $X \sim \mathrm{Binomial}(n, p)$

$E[X] = np \qquad \mathrm{Var}(X) = np(1-p)$

(3) $X \sim \mathrm{Poisson}(\lambda)$

$E[X] = \lambda \qquad \mathrm{Var}(X) = \lambda$

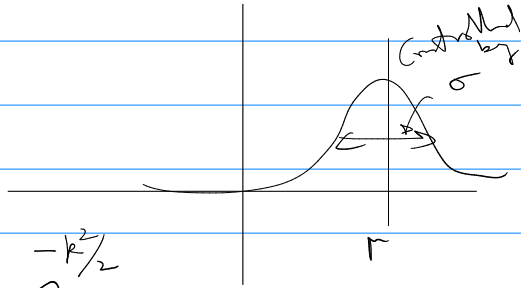
(4) $X \sim \mathrm{Uniform}[a, b]$

$E[X] = \frac{a+b}{2} \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$

(5) $X \sim$ pdf $\frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$

$E[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2$

if $x - \mu = k\sigma$, density $\propto e^{-k^2/2}$



Properties ① $Var(a+X) = Var(X)$

② $Var(aX) = a^2 Var(X)$

③ $X_1, X_2 \sim$ joint distribution

$Var(X_1+X_2) = ?$

$E[(X_1+X_2)^2] = E[X_1^2] + E[X_2^2] + 2E[X_1X_2]$?

$Var(X_1+X_2) = E[X_1^2] + E[X_2^2] + 2E[X_1X_2]$

$- E[X_1]^2 - E[X_2]^2 - 2E[X_1]E[X_2]$